Name: _____

Making Some Ionic Compounds

A. Heat some magnesium ribbon over a portable Bunsen. Don't look at the flames

1. What colour was the material produced?

   _____

The material produced is Magnesium oxide.

When the Magnesium burns each magnesium atom loses 2 electrons and forms the ion

Mg^{+2}

The oxygen atoms in the air near the burning magnesium gain two electrons and form the ion

O^{-2}

These ions have different charges (positive and negative) so they will join together by _____ bonds.

2. Write the formula for the magnesium oxide.

Mg^{+2} O^{-2}

MgO

B. Sprinkle some iron filings over the blue heating flame of the portable Bunsen. WEAR SAFETY GLASSES

1. What colour was the material produced?

   _____

The material produced is iron oxide.

When the iron burns each iron atom loses 3 electrons and forms the ion

The oxygen atoms in the air near the burning iron gain two electrons and form the ion

These ions have different charges (positive and negative) so the will bond together to form _____ bonds.

2. Write the formula for the iron oxide.

$$Fe^{+3} \quad 0^{-2}$$

Fe O

C. Heat some fine copper wire over a portable Bunsen. Wear safety glasses.

3. What colour was the material produced?

The material produced is copper oxide.

When the copper burns each copper atom loses 2 electrons and forms the ion

Cu

The oxygen atoms in the air near the burning copper gain two electrons and form the ion

These ions have different charges(positive and negative) so the will bond together to form _____ bonds.

4. Write the formula for the copper oxide.